

ET4351 Digital VLSI Systems on Chip  
Project Report — Academic Year 2025/2026

Group **XX**

Team Members & Individual Contributions

| #  | First Name | Last Name | Student ID | Individual Contribution |
|----|------------|-----------|------------|-------------------------|
| 1  |            |           |            |                         |
| 2  |            |           |            |                         |
| 3  |            |           |            |                         |
| 4  |            |           |            |                         |
| 5  |            |           |            |                         |
| 6  |            |           |            |                         |
| 7  |            |           |            |                         |
| 8  |            |           |            |                         |
| 9  |            |           |            |                         |
| 10 |            |           |            |                         |

“Primary Responsibility”: briefly state each member’s main contribution (e.g., “Butterfly unit RTL”, “Genus synthesis & clock gating”, “Post-layout sim & VCD”).

Report Guidelines & Knock-Out Criteria

- 1) **Format:** IEEE double-column, compiled to PDF. Font size: 10pt (IEEE default). Use this template.
- 2) **Page limit:** The main body (Sections I–IV) must fit within **6 pages** ( $\approx 3$  pages per design). This cover page and all appendices are **excluded** from the page count.
- 3) **Structure:** You **must** follow the mandatory section structure defined in this template. Do not add, remove, or rename top-level sections (I–IV).
- 4) **Two designs:** Present both a **High-Performance (HP)** and an **Energy-Efficient (EE)** accelerator. Shared aspects should be described once; per-design subsections are provided. **Targets at 12 MHz:** HP — **latency**  $< 61.00\ \mu\text{s}$  (baseline); EE — **accelerator power**  $< 0.403\ \text{mW}$  (baseline). Clock period is **free to change**.
- 5) **Submission checklist** (all mandatory):
  - ☐ Report (PDF) with cover page filled in
  - ☐ SystemVerilog/Verilog source files (cleaned & commented)
  - ☐ All synthesis and P&R scripts
  - ☐ finaldesign\_hp/ and finaldesign\_ee/ directories (each with accel\_audio.hex, et4351.phys.sdf, et4351.phys.v)
  - ☐ All signoff reports: included as files in the project zip **and** as screenshots/excerpts in the PDF appendix
  - ☐ Activity annotation and full power reports: included as files in the project zip **and** as screenshots/excerpts in the PDF appendix
- 6) **Signoff requirements** (per design): core area  $596.4\ \mu\text{m} \times 596.4\ \mu\text{m}$ ; setup & hold timing clean, fully DRV clean (max\_tran violations are acceptable but risky near zero slack); connectivity, geometry, antenna reports clean; functionally correct FFT on variable-length audio input.
- 7) **Quality:** Proper use of technical terms. Reasonably proofread. Clean, readable diagrams.
- 8) **Deadline:** Friday, April 10, 2026 at 16:59 CET.

# ET4351 Project Report — Group **XX**

## I. HIGH-PERFORMANCE (HP) ACCELERATOR

### A. Architecture and Design Methodology

Your content here.

### B. Implementation

Your content here.

### C. Performance Metrics and Validation

Your content here.

TABLE I

EXAMPLE POST-LAYOUT RESULTS (PLEASE IMPROVE OR ORGANIZE IN YOUR OWN WAY)

| Metric                            | Value                                            |
|-----------------------------------|--------------------------------------------------|
| <i>Timing &amp; Area</i>          |                                                  |
| Frequency                         | XXX MHz                                          |
| Core Area                         | 596.4 $\mu\text{m}$ $\times$ 596.4 $\mu\text{m}$ |
| Latency (cycles)                  | XXX                                              |
| Latency ( $\mu\text{s}$ )         | XXX                                              |
| <i>Sign-off</i>                   |                                                  |
| Setup WNS (ns)                    | XXX                                              |
| Hold WNS (ns)                     | XXX                                              |
| DRC Violations                    | XXX                                              |
| DRV Violations                    | XXX                                              |
| <i>Power (activity-annotated)</i> |                                                  |
| Dynamic Power                     | XXX mW                                           |
| Leakage Power                     | XXX mW                                           |
| Total Power                       | XXX mW                                           |
| Energy ( $P \times T$ )           | XXX nJ                                           |
| Annotation Coverage               | XXX %                                            |

TABLE II

EXAMPLE PPA COMPARISON (PLEASE IMPROVE OR ORGANIZE IN YOUR OWN WAY)

| Metric                           | Baseline           | HP                 | EE                 |
|----------------------------------|--------------------|--------------------|--------------------|
| Frequency (MHz)                  | 12                 | —                  | —                  |
| Core Area ( $\mu\text{m}^2$ )    | 596.4 <sup>2</sup> | 596.4 <sup>2</sup> | 596.4 <sup>2</sup> |
| Latency (cycles)                 | 732                | —                  | —                  |
| Latency ( $\mu\text{s}$ )        | 61.00              | —                  | —                  |
| Accel. Dynamic Power (mW)        | 0.372              | —                  | —                  |
| Accel. Leakage Power (mW)        | 0.013              | —                  | —                  |
| Accel. Power (excl. RISC-V) (mW) | 0.403              | —                  | —                  |
| Total Power (mW)                 | 0.626              | —                  | —                  |
| Energy (nJ)                      | 2.46               | —                  | —                  |
| Setup WNS (ns)                   | —                  | —                  | —                  |
| Hold WNS (ns)                    | —                  | —                  | —                  |
| DRC Violations                   | —                  | —                  | —                  |
| Annotation Coverage (%)          | —                  | —                  | —                  |

## II. ENERGY-EFFICIENT (EE) ACCELERATOR

### A. Architecture and Design Methodology

Your content here.

### B. Implementation

Your content here.

### C. Performance Metrics and Validation

Your content here, including a table similar to Table I.

## III. COMPARATIVE PPA TRADE-OFF ANALYSIS

You can discuss the trade-offs revealed by Table II and any other relevant observations. For example:

- Which architectural and tool-constraint decisions had the largest impact on each PPA dimension?
- What would you do differently if redesigning?
- Any surprising results or lessons learned.

## REFERENCES

- [1] A. Author, "Paper title," *Journal*, vol. X, pp. Y–Z, Year.
- [2] B. Author, *Book Title*. City: Publisher, Year.

## APPENDIX

### A. *High-Performance (HP) Design*

- A.1 *Timing Reports:*
- A.2 *DRC Reports:*
- A.3 *DRV Reports:*
- A.4 *Connectivity Reports:*
- A.5 *Geometry Reports:*
- A.6 *Antenna Reports:*
- A.7 *Activity Annotation Reports:*
- A.8 *Power Analysis Reports:*
- A.9 *Sign-off Simulation Waveforms:*

### B. *Energy-Efficient (EE) Design*

- B.1 *Timing Reports:*
- B.2 *DRC Reports:*
- B.3 *DRV Reports:*
- B.4 *Connectivity Reports:*
- B.5 *Geometry Reports:*
- B.6 *Antenna Reports:*
- B.7 *Activity Annotation Reports:*
- B.8 *Power Analysis Reports:*
- B.9 *Sign-off Simulation Waveforms:*